Untitled

Habet Madoyan

August 27, 2018

```
install.packages("caret")
```

```
library(caret)
createDataPartition()
```

```
library(caret)
caret::createDataPartition()
```

```
census <- read.csv("census.csv")
```

```
summary(census)
```

```
##        id             race           region           sex
##  Min.   :   1   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.: 524   1st Qu.:1.000   1st Qu.:2.000   1st Qu.:1.000
##  Median :1015   Median :1.000   Median :4.000   Median :2.000
##  Mean   :1019   Mean   :1.309   Mean   :4.085   Mean   :1.546
##  3rd Qu.:1530   3rd Qu.:1.000   3rd Qu.:6.000   3rd Qu.:2.000
##  Max.   :2023   Max.   :3.000   Max.   :8.000   Max.   :2.000
##      degree           sibs           wrkstat         childs
##  Min.   :0.000   Min.   : 0.000   Min.   :1.00   Min.   :0.000
##  1st Qu.:1.000   1st Qu.: 2.000   1st Qu.:1.00   1st Qu.:0.000
##  Median :1.000   Median : 3.000   Median :2.00   Median :2.000
##  Mean   :1.565   Mean   : 3.775   Mean   :2.97   Mean   :1.955
##  3rd Qu.:3.000   3rd Qu.: 5.000   3rd Qu.:5.00   3rd Qu.:3.000
##  Max.   :8.000   Max.   :99.000   Max.   :8.00   Max.   :9.000
##      hompop           earnrs          adults           educ
##  Min.   : 1.000   Min.   :0.000   Min.   :1.000   Min.   : 0.00
##  1st Qu.: 1.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:12.00
##  Median : 2.000   Median :1.000   Median :2.000   Median :13.00
##  Mean   : 2.519   Mean   :1.453   Mean   :1.883   Mean   :13.58
##  3rd Qu.: 3.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:16.00
##  Max.   :11.000   Max.   :9.000   Max.   :9.000   Max.   :99.00
##     marital          happy           income           age
##  Min.   :1.000   Min.   :1.000   Min.   : 1.00   Min.   :18.00
##  1st Qu.:1.000   1st Qu.:1.000   1st Qu.:11.00   1st Qu.:34.00
##  Median :2.000   Median :2.000   Median :12.00   Median :47.00
##  Mean   :2.515   Mean   :1.895   Mean   :14.57   Mean   :47.82
##  3rd Qu.:5.000   3rd Qu.:2.000   3rd Qu.:12.00   3rd Qu.:60.00
##  Max.   :5.000   Max.   :3.000   Max.   :98.00   Max.   :99.00
##       sei             maeduc          pasei          agekdbrn
##  Min.   :-1.00   Min.   : 0.00   Min.   :-1.00   Min.   : 0
##  1st Qu.:31.20   1st Qu.:11.00   1st Qu.:26.50   1st Qu.: 0
##  Median :38.40   Median :12.00   Median :36.40   Median :21
##  Mean   :45.82   Mean   :21.81   Mean   :36.12   Mean   :18
##  3rd Qu.:63.50   3rd Qu.:15.00   3rd Qu.:50.70   3rd Qu.:25
##  Max.   :97.20   Max.   :99.00   Max.   :97.20   Max.   :99
##      paeduc          hapunhap         trust
##  Min.   : 0.00   Min.   :0.000   Min.   :0.000
##  1st Qu.:11.00   1st Qu.:0.000   1st Qu.:0.000
##  Median :12.00   Median :1.000   Median :1.000
##  Mean   :34.22   Mean   :1.137   Mean   :1.165
##  3rd Qu.:97.00   3rd Qu.:2.000   3rd Qu.:2.000
##  Max.   :99.00   Max.   :4.000   Max.   :3.000
```

```r
census$happy<-factor(census$happy, levels=c(1,2,3),
                     labels=c("Very Happy", "Pretty Happy", "Not too Happy"))
```

```r
str(census)
```

```
## 'data.frame':    1929 obs. of  23 variables:
##  $ id      : int  1 2 3 4 6 11 12 14 17 19 ...
##  $ race    : int  3 3 2 2 2 2 2 2 1 1 ...
##  $ region  : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ sex     : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ degree  : int  1 4 1 1 3 4 4 1 1 3 ...
##  $ sibs    : int  0 1 1 2 3 4 8 7 3 0 ...
##  $ wrkstat : int  8 1 1 4 5 1 1 5 1 1 ...
##  $ childs  : int  5 2 1 1 2 2 0 4 0 0 ...
##  $ hompop  : int  2 1 2 3 2 2 1 2 2 1 ...
##  $ earnrs  : int  0 2 1 2 2 2 1 0 2 1 ...
##  $ adults  : int  1 1 2 3 2 2 1 2 2 1 ...
##  $ educ    : int  12 20 13 10 16 17 19 16 15 16 ...
##  $ marital : int  2 1 1 3 1 1 5 1 5 5 ...
##  $ happy   : Factor w/ 3 levels "Very Happy","Pretty Happy",..: 3 2 2 3 1 2 3 1 2 2 ...
##  $ income  : int  7 12 12 12 13 12 12 12 12 12 ...
##  $ age     : int  49 48 47 32 72 62 40 71 26 36 ...
##  $ sei     : num  28.4 76.4 37.8 35.1 87.9 73.9 73.3 63.2 54.5 81 ...
##  $ maeduc  : int  12 12 16 12 12 12 8 15 12 12 ...
##  $ pasei   : num  28.4 -1 -1 -1 36.4 27.7 36.4 45.9 84.7 63.5 ...
##  $ agekdbrn: int  20 33 22 26 25 31 0 22 0 0 ...
##  $ paeduc  : int  12 97 97 97 98 6 8 12 16 12 ...
##  $ hapunhap: int  2 0 2 2 1 1 3 1 0 2 ...
##  $ trust   : int  2 0 2 1 2 1 2 2 0 2 ...
```

```r
table(census$happy)
```

```
##
##    Very Happy  Pretty Happy Not too Happy
##           516          1099           314
```

```r
census8 <- census[census$region==8,]
table(census8$happy)
```

```
##
##    Very Happy  Pretty Happy Not too Happy
##             0           178            54
```

```r
census$happy <- as.character(census$happy)
census8 <- census[census$region==8,]
table(census8$happy)
```

```
##
## Not too Happy  Pretty Happy
##            54           178
```

```r
x = 10
x == 10
```

```
## [1] TRUE
```

```r
save(census, file='census.rda')
```

```r
rm(list=ls())
```

```r
load('census.rda')
```